

(A) List

```
a = [1, 2, 3]
a[0] = 10
print(a)
```

What is the output?

- A. Error
- B. [10, 2, 3]
- C. [1, 2, 3]
- D. None

(B) Tuple

```
a = (1, 2, 3)
a[0] = 10
```

What happens?

- A. (10, 2, 3)
- B. No change
- C. Error
- D. None

(C) Set

```
a = {1, 2, 3}
a[0] = 10
```

What happens?

- A. Works
- B. Error
- C. Adds element
- D. No change

(A) List

```
a = [1, 1, 2, 3]
print(len(a))
```

(B) Tuple

```
a = (1, 1, 2, 3)
print(len(a))
```

(C) Set

```
a = {1, 1, 2, 3}
print(len(a))
```

(A) List

```
a = [10, 20, 30]  
print(a[1])
```

(B) Tuple

```
a = (10, 20, 30)  
print(a[1])
```

(C) Set

```
a = {10, 20, 30}  
print(a[1])
```

A)List

```
a = [1, 2]
a.append(3)
print(a)
```

(B) Tuple

```
a = (1, 2)
a.append(3)
```

(C) Set

```
a = {1, 2}
a.add(3)
print(a)
```

Which statement is TRUE?

(A) List

A. Ordered

B. Unordered

(B) Tuple

A. Ordered

B. Unordered

(C) Set

A. Ordered

B. Unordered

(A)

```
a = (5)
print(type(a))
```

(B)

```
a = (5,)
print(type(a))
```

(C)

```
a = {5}
print(type(a))
```

A) List

```
a = [1, 2, 3]  
a.remove(4)
```

(B) Tuple

```
a = (1, 2, 3)  
a.remove(2)
```

(C) Set

```
a = {1, 2, 3}  
a.remove(4)
```

(A) List

```
a = [1, 2, 3]
print(a[0:2])
```

(B) Tuple

```
a = (1, 2, 3)
print(a[0:2])
```

(C) Set

```
a = {1, 2, 3}
print(a[0:2])
```


(A) List

```
a = [1, [2, 3]]  
a[1][0] = 99  
print(a)
```

(B) Tuple

```
a = (1, [2, 3])  
a[1][0] = 99  
print(a)
```

(C) Set

```
a = {1, [2, 3]}  
print(a)
```

(A) List

```
a = [1, 2] + [3, 4]  
print(a)
```

(B) Tuple

```
a = (1, 2) + (3, 4)  
print(a)
```

(C) Set

```
a = {1, 2} + {3, 4}  
print(a)
```

(A) List

```
a = [1, 2] * 2  
print(a)
```

(B) Tuple

```
a = (1, 2) * 2  
print(a)
```

(C) Set

```
a = {1, 2} * 2  
print(a)
```

A)

```
list((1, 2, 3))
```

(B)

```
tuple([1, 2, 3])
```

(C)

```
set([1, 2, 2, 3])
```

(A) List

```
a = [3, 1, 2]  
a.sort()
```

(B) Tuple

```
a = (3, 1, 2)  
a.sort()
```

(C) Set

```
a = {3, 1, 2}  
a.sort()
```

(A) List

```
a = [1, 2, 3]  
a.index(2)
```

(B) Tuple

```
a = (1, 2, 3)  
a.index(2)
```

(C) Set

```
a = {1, 2, 3}  
a.index(2)
```

Match each method with its correct function:

Column A (Method / Function / Keyword)	Column B (Purpose)
append()	A. Combines all elements of sets
add()	B. Returns sorted copy without modifying original
extend()	C. Adds single element to set
insert()	D. Removes element safely without error
remove()	E. Adds multiple elements from iterable
discard()	F. Inserts element at a specific position
pop()	G. Removes first occurrence of value
sort()	H. Adds element at end of list (in-place)
sorted()	I. Removes and returns last element
index()	J. Finds position of element
count()	K. Counts occurrences of element
union()	L. Removes duplicate elements automatically (concept)
intersection()	M. Finds common elements between sets
difference()	N. Finds elements present in first set but not second
symmetric_difference()	O. Finds non-common elements
clear()	P. Removes all elements
len()	Q. Returns number of elements
sum()	R. Returns total of numeric elements
min()	S. Returns smallest element
max()	T. Returns largest element
copy()	U. Creates shallow copy of object
del	V. Deletes object or element from memory
reversed()	W. Returns reverse iterator

Each group must find the odd one out and explain why:

Set 1:

- append()
- extend()
- add()
- insert()

Set 2:

- union()
- intersection()
- sort()
- difference()

Set 3:

- len()
- max()
- append()
- sum()

Set 4:

- add()
- remove()
- discard()
- index()

Set 5:

- list
- tuple
- set
- integer

****Set 6**

- sorted()
- reversed()
- sort()
- len()

Set 7

- list()
- tuple()
- set()
- append()